MSCF Probability – Final Exam – August 17, 2018

Name ____

Email ____

Instructions:

- You should complete all questions on this exam by writing out the solutions on paper that will be provided to you. Anything written on the exam paper itself will **not** be graded.
- You will have three hours to complete this exam. No matter when you start the exam, you have to stop at 3:00 PM.
- Notes (except described below), books and calculators/laptops/phones are not allowed.
- A table of common distributions, with pdfs/pmfs, and other information, is included at the end of this exam.
- Final answers can involve any of the following notation:

$$\binom{n}{k},\ n^k,\ \exp(k),\ n!,\ \Gamma(x),\ \Phi(x),$$

  where $\Phi(x)$ is the CDF of the standard normal distribution. Except for cases in which it is explicitly stated, final answers should **not** include integrals.
- You may use 2 handwritten sides of a 8.5" by 11" page (1 double-sided or 2 single-sided) of notes.
- Questions are not all worth the same number of points. The numbers in **bold** after each question indicate the number of points. Also, you may find some easier than others. You should prioritize your work on particular questions accordingly.
  Note: **The total number of points is XXX.**
- You **must show all of your work**, unless otherwise stated. A correct answer without explanation will receive only minimal or no partial credit. Note that the "easier" a question may seem, the more burden you have to show that you understand how you are reaching the answer; again, **be clear and complete in your work**.

PLEASE CHECK THAT YOU HAVE PAGES 1 – XXX, THE DISTRIBUTION TABLE, AND THE MATH FACTS

Grading Guidelines:

The following are reasons why you may lose points:

1. Failure to mention when an independence assumption is utilized.
2. Failure to mention when the assumption that random variables are uncorrelated is utilized.
3. Failure to specify clearly the support of a distribution.

You do **not** need to cite the names of rules and definitions when you use them, such as

1. The Rule for the Lazy Statistician
2. The Iterated Expectation Rule
3. The Rule for Conditional Probability
4. The Addition Rule
5. The Chain Rule
6. ...

But, in general, any work that does not make it clear how the answer was found is subject to losing points. In other words, even if you do not justify every step, there should be sufficient number of steps to make it clear you know how to do the derivation. Lack of clarity can also result from messy writing: write neatly.